Project Design Phase
Problem – Solution Fit Template

Date	15 February 2025
Team ID	LTVIP2025TMID36086
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	2 Marks

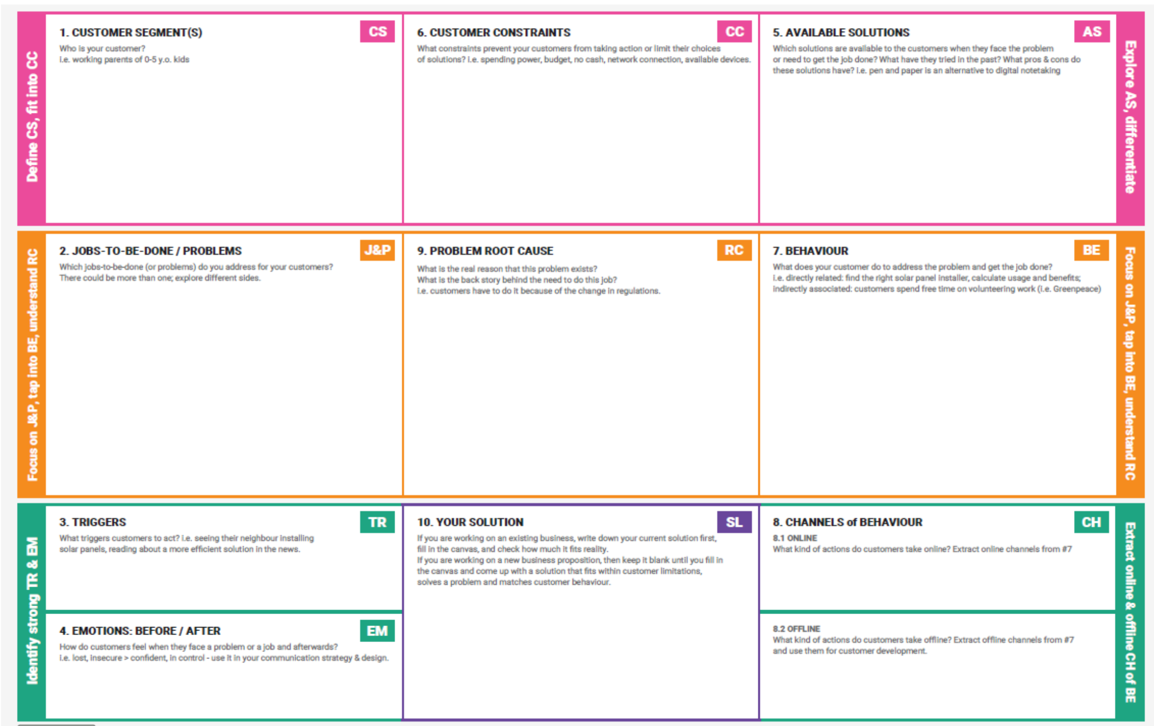
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

🔍 Purpose of the Canvas

- Solve a real problem experienced by biodiversity researchers, ecologists, and citizen scientists.
- Increase adoption by building an intuitive, AI-powered butterfly identification tool that aligns with user behaviors.
- Improve engagement through environmental education and science communication.
- Build trust with stakeholders by solving a recurring field problem: species misidentification.

🧠 Customer Segment / User Persona

- Field researchers involved in biodiversity tracking
- Students and educators in environmental sciences
- Nature photographers and citizen scientists

❗ The Problem (User Pain Points)

- Manual butterfly identification is time-consuming and error-prone.
- Non-experts (e.g., students, public) lack tools for accurate species recognition.
- Existing apps are either outdated, inaccurate, or not inclusive of diverse species (like 75 classes).
- Researchers lose time cross-referencing field guides or expert opinions.

💡 The Solution

- A transfer learning-based image classification system that accurately identifies butterfly species from images.
- Fast, mobile-friendly predictions using a simple web interface (Streamlit-based).

- Trained on 6499+ images across 75 butterfly classes to ensure generalization.
- Real-time accuracy, visualization, and classification feedback for both experts and beginners.

🛠️ Key Features

- VGG16 or ResNet-based transfer learning architecture
- Model accuracy ~93% with fine-tuning
- Streamlit or Flask frontend for user interaction
- Educational output with species name, confidence score, and ecological relevance

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🧩 Problem–Solution Fit

Component	Details
User Behavior	Captures butterflies, wants instant ID, contributes to science
Problem Encountered	No accurate, fast, or field-friendly classification tool available
Solution Offered	AI model that works offline/online and simplifies classification
Behavior Match	Users already use cameras, mobile tools – the solution extends their habit
Value Proposition	Saves time, increases ID accuracy, supports conservation with meaningful data

Problem–Solution Fit Template

Project: Enchanted Wings: Marvels of Butterfly Species

Team ID: LTVIP2025TMID668

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Key Features

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